Geophys. J. Int. (2008) **173,** 63–78 doi: 10.1111/j.1365-246X.2007.03687.x

Palaeomagnetic confirmation of Palaeozoic clockwise rotation of the Famatina Ranges (NW Argentina): implications for the evolution of the SW margin of Gondwana

C. M. Spagnuolo,[1] A. E. Rapalini[1] and R. A. Astini[2]

[1]*INGEODAV, Depto.Cs.Geológicas, F.C.E.yN., Universidad de Buenos Aires, Pabellón 2, Ciudad Universitaria,* C1428EHA*, Buenos Aires, Argentina and CONICET. E-mail: cspagnuolo@gl.fcen.uba.ar*

[2]*Laboratorio de Análisis de Cuencas, Cátedra de Estratigrafía y Geología Histórica, F.C.E.F.y N., Universidad Nacional de Córdoba,Av. Velez Sarsfield* 1611*, X5016GCA, Córdoba, Argentina and CONICET*

Accepted 2007 November 8. Received 2007 November 7; in original form 2007 April 11

SUMMARY

Palaeomagnetic results from Palaeozoic volcanic and sedimentary units of the Famatina Ranges, in NW Argentina (28.7°S, 67.8°W) are reported. A late Early to late Middle Ordovician palaeomagnetic pole was obtained from a pre-tectonic remanence carried by magnetite and isolated in volcanics of the Molles Formation and the Cerro Morado Group (MCM1, 16.7°S, 357.2°E, $A_{95} = 6.5°$, $K = 38.5$, $N = 14$ sites). This pole position is rotated 39° clockwise respect to the coeval reference pole for Gondwana but it is consistent with previous Early Ordovician poles from the Famatina belt and the Faja Eruptiva Oriental in the Puna region of NW Argentina. The sedimentary layers of the Molles Formation, however, present a secondary magnetization carried by hematite, which is interpreted of Permian age and yields a pole position (MCM2) at 78.7°S, 330.8°E ($A_{95} = 7.2°$, $K = 16.1$, $n = 27$ samples). Two additional independent palaeomagnetic poles were obtained from the Permian De La Cuesta Formation, exposed at two different localities in the same area. While one consisted in a exclusively reverse polarity magnetization and a pole position (LC1, 76.9°S, 345.2°E, $A_{95} = 6.0°$, $K = 21.1$, $n =$ 29 samples) compatible with the late Early to early Late Permian palaeomagnetic poles from South America, the other presented only normal polarities and a pole position (LC2, 74.5°N, 275.4°E, $A_{95} = 2.0°$, $K = 258.3$, $n = 21$ samples) suggestive of a Cretaceous remagnetization. These new palaeomagnetic results confirm on a much more robust database previous proposals that the Ordovician rocks of the Famatina belt have undergone a large clockwise rotation. They also constrain the rotation to pre-Permian times. Different tectonic models involving the Late Ordovician docking of a large para-authochthonous terrane or a pattern of systematic large-scale rotations in the Early Palaeozoic continental margin of Western Gondwana are discussed.

Key words: Palaeomagnetism applied to tectonics; Remagnetization; Rock and mineral magnetism; Continental margins: convergent; Volcaniclastic deposits; South America.

1 INTRODUCTION

The Famatina Ranges extend north–south for more than 400 km in NW Argentina, reaching over 6000 m above sea level (Fig. 1A). A thick and well-exposed Ordovician rock succession makes it a key area for the study of the SW continental margin of Gondwana in the Early Palaeozoic (see Pankhurst & Rapela 1998; Astini & Dávila 2004, and references therein). The supracrustal Ordovician units are composed of more than 3200 m of rhyolitic to basaltic lavas, volcano-sedimentary deposits, carbonates and siliciclastic rocks (Astini 2003). These rocks have been interpreted as extruded and deposited within a volcanic-arc environment. Huge outcrops of Early Ordovician granitoids also extend along most of the Famatina belt as well as in many other localities of the Sierras Pampeanas in western Argentina (e.g. Llambías *et al.* 1998; Pankhurst & Rapela 1998). Geochemical signature of both lavas and granitoids suggests the magmatic arc developed on continental crust along the SW margin of Gondwana during the Early Palaeozoic (Pankhurst *et al.* 2000). According to recent interpretations this continental arc may have extended down to NE Patagonia, for nearly 1500 km (Pankhurst *et al.* 2006). Cessation of magmatism by late Middle to Late Ordovician has been interpreted as evidence for subduction demise after collision of the Laurentian derived Precordillera terrane (Thomas & Astini 2003; Ramos 2004). Previous palaeomagnetic studies in Ordovician rocks of the Famatina Ranges (Valencio *et al.* 1980; Conti *et al.* 1996) and coeval igneous and sedimentary

© 2008 The Authors
Journal compilation © 2008 RAS

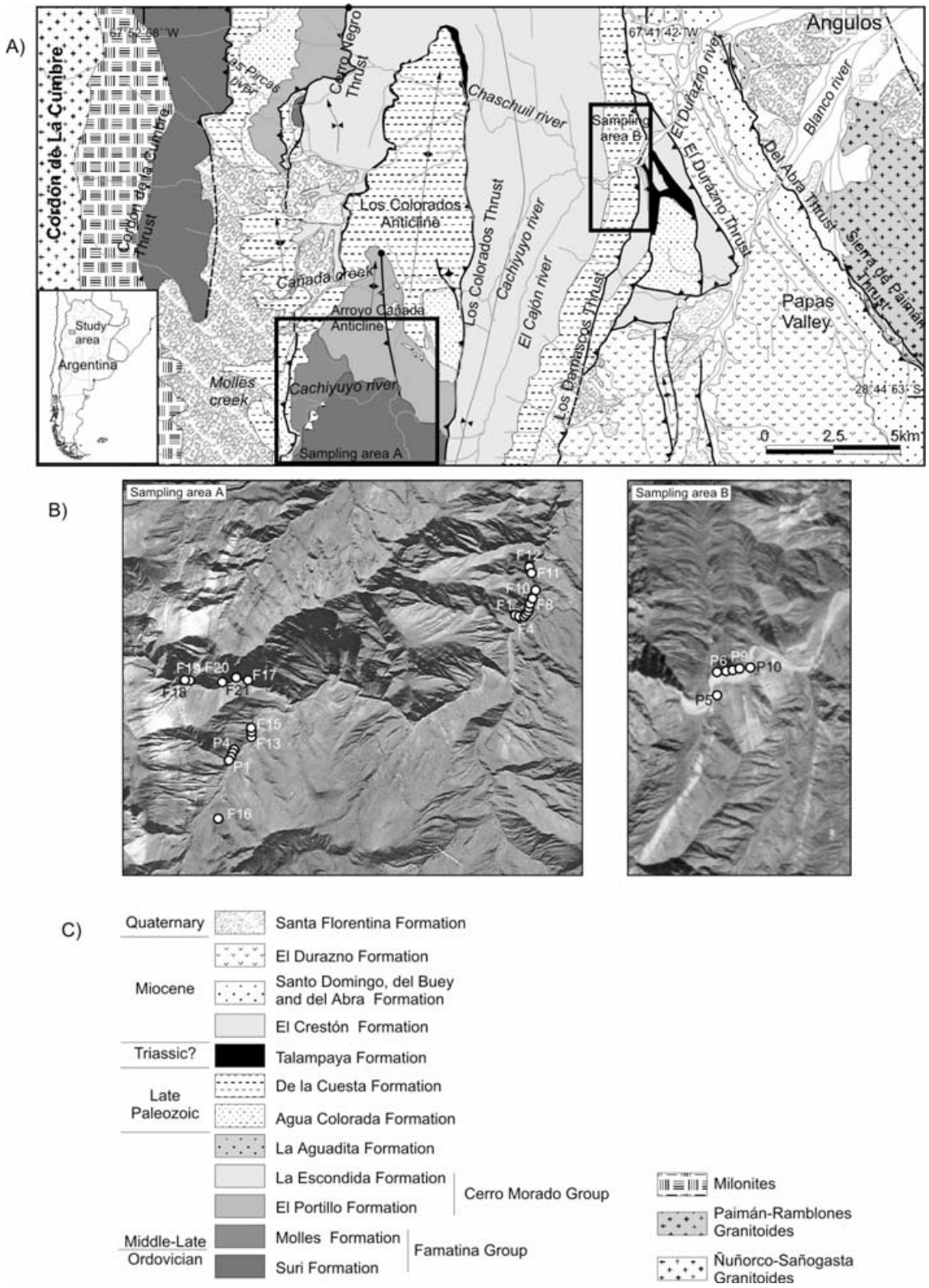


Figure 1. (A) Location of the Famatina Range in South America. (B) Satellite image of the study area and location of the sampling sites. (C) References of the geological map (taken from Dávila 2003).

rocks in the Eastern Puna in NW Argentina (Conti *et al.* 1996) were interpreted as evidence for a large para-authocthonous terrane, that rotated during the Middle (or Late?) Ordovician prior to or during the collision of Precordillera. This rotation implies closure of an ocean-floored(?) backarc basin to the East of this terrane. This model is compatible with some biogeographic (Benedetto 1998), magmatic (Quenardelle & Ramos 1999) and geophysical (Martínez & Giménez 2003) evidence. The presence of basic metamorphic

rocks interpreted as remnants of a backarc (Las Termas belt, Miller & Sollner 2005), immediately to the East of the Famatina arc is also consistent with this model, which was first postulated by Conti *et al.* (1996). However, this has been disputed by several authors (Pankhurst *et al.* 1998; Rapela *et al.* 1998; Saavedra *et al.* 1998; Astini 2003) in favour of a model of an Andean-type continental magmatic arc developed on the Gondwana margin. This alternative model is mainly based upon the geochemical signature of the magmatic products that are not compatible with an intraoceanic tectonic setting (see Rapalini 2005 for a recent brief review on the subject). The palaeomagnetic data upon which the first model was based was not without limitations. First, no conclusive evidence for the time of rotation was available at each locality, either for the Famatina belt or the Eastern Puna. Secondly, the conclusion was drawn from four individual palaeomagnetic poles which are very consistent but that were obtained with a moderate to small number of sites and samples (Conti *et al.* 1996). No further palaeomagnetic studies had been conducted on Palaeozoic rocks of the Famatina belt or in Puna since then. Therefore, a more systematic palaeomagnetic study was carried out on a larger number of sites in the central area of the Sierra de Famatina on stratigraphically better constrained larger number of sites and units of different ages

2 GEOLOGICAL SETTING AND PALAEOMAGNETIC SAMPLING

The sampling area is located on the eastern slope of the Famatina Range in the north of the La Rioja Province (Fig. 1A), between 28°44.4 S–28°41.1 S and 67°50.0 W–67°43.6 W. The stratigraphy of the region is relatively simple (Turner 1964; Astini & Dávila 2004) comprising a Late Proterozoic to Early Cambrian metamorphic basement, Early to Middle Ordovician volcanics and sedimentary units, Late Palaeozoic strata, and a Cenozoic sedimentary cover (Fig. 1C).

The structure of the region is dominated by an East-vergent fold and thrust belt (Fig. 1). Sampling localities (Fig. 1B) were determined on the conspicuous Los Colorados anticline with a north trending (8°E) and gently plunging to the north (16°) axis (Dávila *et al.* 2003; Astini & Dávila 2004), the Arroyo Cañada anticline (350° north trending axis plunging 33° towards the north) and the Los Damascos thrust, located to the East.

Palaeomagnetic sampling was carried out on the late Early to early Middle Ordovician Molles Formation, the Middle Ordovician Cerro Morado Group and the Permian De La Cuesta Formation.

The Molles Formation (Harrington 1957), the uppermost unit of the Famatina Group, is composed of volcanogenic sandstones, breccias, tuffs and some intercalated basic lavas (Turner 1964; Astini 2003) that have been assigned to an intraarc to interarc setting (Astini 2003; Astini & Dávila 2004). According to conodonts and acritarcs associations, an early Middle Arenig to late Middle Arenig age has been assigned to this unit (see Astini 2003 and references in there).

The late Middle Ordovician Cerro Morado Group (Astini & Dávila 2002) overlies unconformably the Famatina Group. It comprises the El Portillo and La Escondida Formations. El Portillo Formation is 580 m thick and composed of lava flows (rhyolites and rhyodacites) and interbedded ignimbrites and tuffs (Astini 2003). La Escondida Formation includes ignimbrites, tuffs and volcanogenic sandstones, lapilli layers and few shales.

The Permian deposits lie unconformably on the Ordovician rocks. The De La Cuesta Formation (Turner 1960) is composed of bright red fluvial and ephemeral stream facies that gradually pass to reddish eolian sandstones and lacustrine facies towards the top (Dávila 2003). López Gamundi *et al.* (1994) proposed a Permian age to this unit according to palaeontological studies.

Two hundred and eight oriented cores were collected from thirty-one sites with a portable gas-powered drill. Sites and samples were distributed in this way: 13 sites (92 samples) were located on the Cerro Morado rhyolitic/ignimbritic flows; 7 sites (48 samples) belonged to the Molles Formation, one site on rhyolitic and two on basic flows, while the remaining four corresponded to fine-grained red sandstones and siltstones. Eight samples (1 site) were collected from a basic dyke that crosscut the Molles Formation with a possible but uncertain Silurian-Devonian age. Besides this, four sites (25 samples) were located on fine-grained red sandstones of the Permian De La Cuesta Formation. All the above mentioned sampling was located in the Los Colorados Anticline. Further 6 sites (35 cores) were located on outcrops of the De La Cuesta Formation along the El Durazno river (Fig. 1).

Samples were oriented by means of both sun and magnetic compasses. At the laboratory each core was sliced into one to three standard palaeomagnetic specimens (2.2 cm height and 2.54 cm diameter).

Two specimens from each Ordovician site were submitted to a pilot standard stepwise demagnetization procedure, one to thermal and one to AF demagnetization in sixteen steps up to temperatures (fields) of 700 °C (100 mT). This was to evaluate the best demagnetization technique to be applied to the remaining samples. The remaining specimens of the Ordovician rocks were submitted to detailed stepwise thermal demagnetization at 150, 300, 400, 450, 500, 530, 560, 580, 630, 650, 680 and 700 °C (Fig. 2).

Three specimens per site of the Permian rocks were submitted to a pilot standard stepwise demagnetization procedure: one specimen to AF and 2 specimens to thermal demagnetization with the same purposes. The remaining specimens were submitted to detailed stepwise thermal demagnetization at 150, 320, 420, 500, 550, 600, 640, 660, 680 and 700 °C (Fig. 2).

Intensity and direction of the natural remanent magnetization were measured with a DC-SQUID (2G-750R) cryogenic magnetometer. AF cleaning was achieved by means of a static three-axes degausser attached to the cryogenic magnetometer. Thermal demagnetization was applied with an ASC dual chamber oven, with internal magnetic fields below 5 nT. Bulk magnetic susceptibility was measured with a Bartington MS-2 susceptometer after each thermal step to control possible chemical changes induced by heating of the samples. Magnetic components were determined by principal component analysis (Kirschvink 1980) with maximum angular deviation (MAD) values under 11° (but only 11 samples had MAD larger than 6°) for the Ordovician rocks, under 7° for the basic dyke and under 14.5° for the Permian sedimentary rocks (61 per cent of the samples have a MAD below 7°).

Acquisition of isothermal remanence curves were performed with a pulse magnetizer (ASC Scientific IM-10–30) in order to identify the magnetic carriers at each site. One sample per site was analysed through the sequence: 0, 17, 29, 44, 61, 90, 150, 250, 350, 450, 600, 1000, 1310, 1640, 2300 and 2545 mT.

3 PALAEOMAGNETIC RESULTS

3.1 Ordovician volcanics

The rhyolitic and ignimbritic flows of the Cerro Morado Group and Molles Formation have similar magnetic behaviours.

© 2008 The Authors, *GJI*, **173,** 63–78
Journal compilation © 2008 RAS

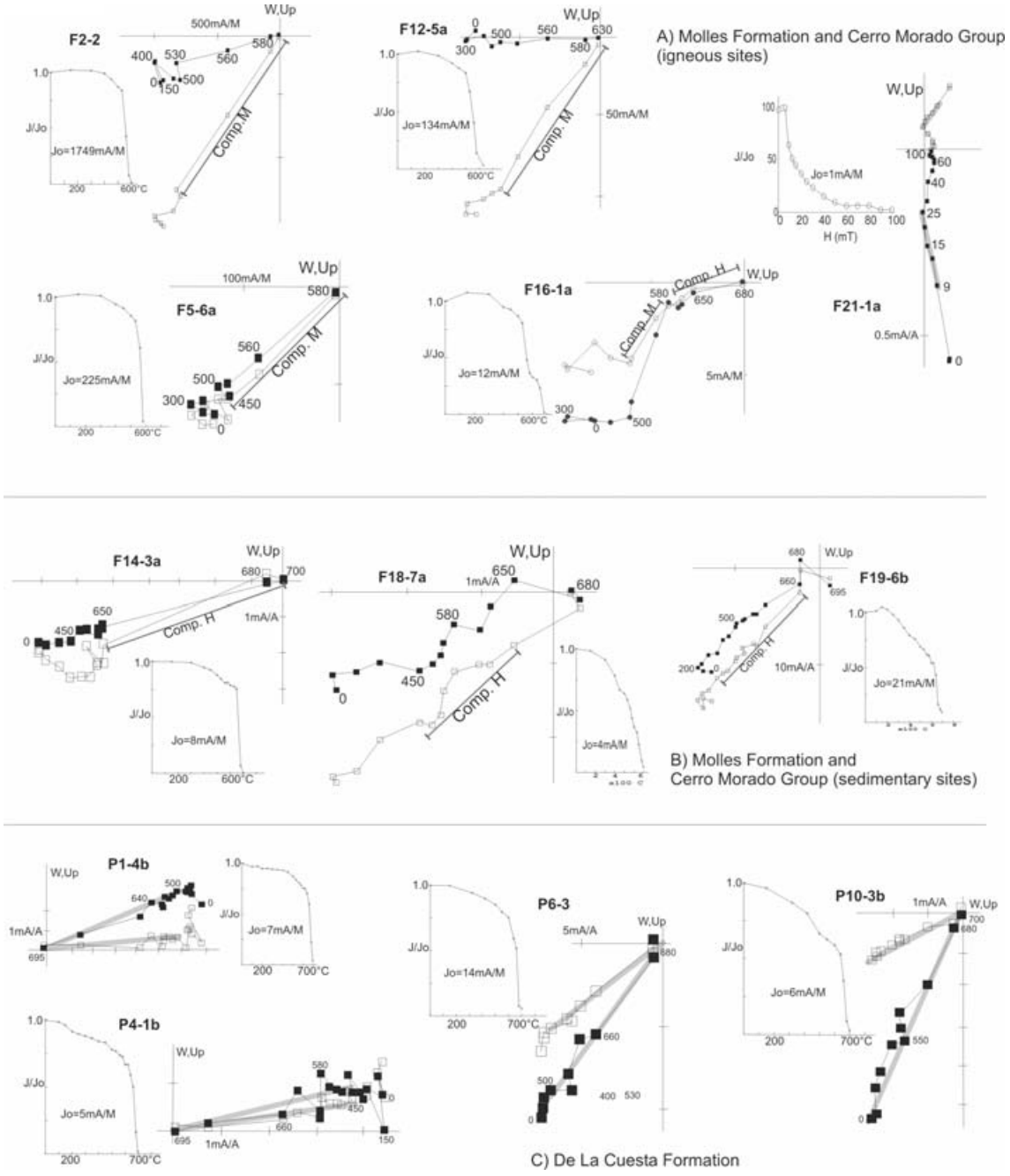


Figure 2. Representative demagnetization behaviour of the analysed samples (geographic coordinates). (A) Igneous samples of the Molles Formation and Cerro Morado Group carrying component M. (B) Sedimentary samples of the Molles Formation and volcanics of the Cerro Morado Group carrying component H. (C) samples from red beds of the De la Cuesta Formation. The magnetic component/s of each sample is/are indicated in the Zijderveld diagrams. Closed (open) symbols represent projections on the horizontal (vertical) plane. More references in the text.

Demagnetization curves (Fig. 2A) show that magnetite is the principal magnetic carrier with unblocking temperatures close to 580 °C. A dominant ferromagnetic phase is also observed in the IRM acquisition curves (Fig. 3A) from all these sites. AF demagnetization shows moderate coercivities (20–80 mT) compatible with the presence of PSD and/or SD grains. A small contribution of (low Ti?) hematite with unblocking temperature of about 630 °C was also observed in ten out of sixteen sites.

A magnetic component of high temperature/high magnetic fields (400–580 °C/15–100 mT) was isolated in most sites and samples (component M). Exception to this was site F-21, correspondent to a pillow lava that presented magnetic minerals of low to medium coercivity (Fig. 2A) and inconsistent remanence directions. Site F-16 presented both component M and a second component (component H) with unblocking temperatures around 680 °C (Fig. 2A) indicating hematite as its magnetic carrier (see also IRM curves, Fig. 3A). The M component of site F-16 is analysed together with those from other sites of the Cerro Morado Group. However component H resembles those from the Ordovician sedimentary rocks and it is analysed with them (see below). Something similar occurred

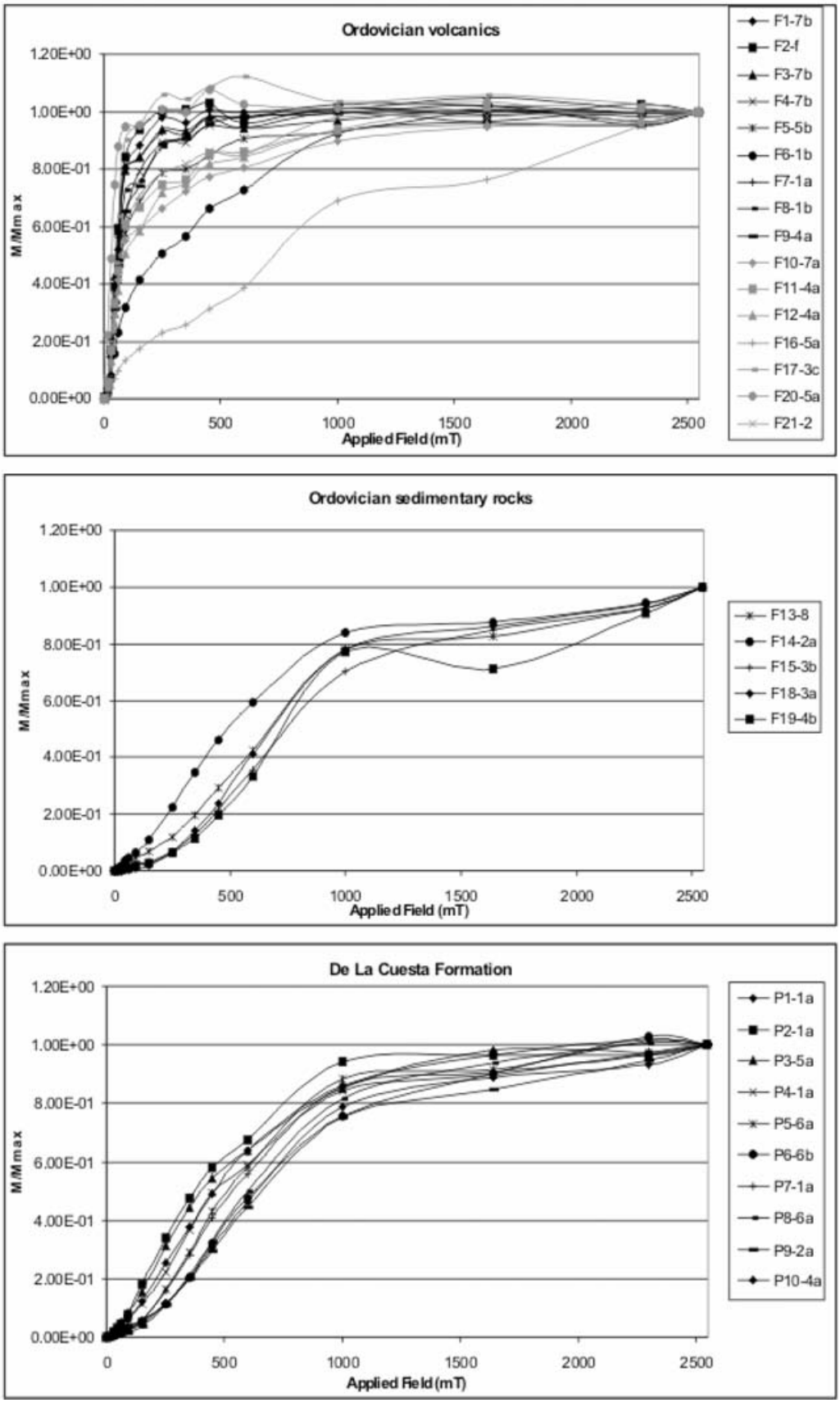


Figure 3. Normalized isothermal remanent magnetization (IRM) acquisition curves. (A) The steep slopes indicate magnetite as the principal magnetic carrier, except for F6–1b and F16–5a in which the presence of both magnetite and hematite is interpreted. (B) and (C) The initial shallow slopes indicate hematite as the principal magnetic carrier.

© 2008 The Authors, *GJI*, **173,** 63–78
Journal compilation © 2008 RAS

to site F-13, an ignimbrite flow that shows magnetic characteristics similar to the sedimentary rocks, with a single magnetic component (H) that is analysed with that group too. Table 1 presents the mean site remanence directions, both before and after the application of the structural correction, for volcanics of the Cerro Morado Group and Molles Formation carriers of component M. Most sites (12 out of 14) presented high directional consistency ($\alpha_{95} < 10^\circ$). Application of bedding correction followed two steps, 'unplunging' of the fold, followed by rotation of the 'unplunged' bedding plane to horizontal. This procedure should eliminate any apparent tectonic rotation (MacDonald 1980). Comparison of mean site remanence directions before and after bedding correction indicates a significant grouping of directions after correction. This can be tested statistically by applying the stringent fold test of Watson & Enkin (1993). After the application of the bedding correction the statistical parameters of the average of site directions improved significantly: *kappa* (K) increases from 8.7 to 48.5 and α_{95} diminishes from 14.3° to 5.8° (Table 1; Fig. 4A). This results in a positive fold test indicating that Component M (the remanence carried by most Ordovician volcanics) is pre-tectonic. The mean of the bedding corrected site remanence is Dec = 93.4°; Inc = 47.7°; N (number of sites) = 14; $\alpha_{95} = 5.8^\circ$ (Fig. 4A). A VGP was calculated for each single site, then a mean VGP was calculated from them, which constitutes the palaeomagnetic pole MCM1: 16.7°S, 357.2°E, $A_{95} = 6.5^\circ$, $K = 38.5$, $N = 14$.

3.2 Ordovician sedimentary rocks

The IRM acquisition curves (Fig. 3B) and the unblocking temperatures (680 °C) of the red sandstones and siltstones of the Molles Formation indicate that hematite is the principal magnetic carrier (Fig. 2B). As usual in these cases AF demagnetization was ineffective. A high temperature component (normally defined in the range 300–700 °C) was isolated in most samples (component H; Table 2). As already mentioned, one rhyolitic site from the Molles Formation (F-13) was analysed together with these rocks due to very similar demagnetization behaviours. Furthermore, site F-16 (rhyolite from the Cerro Morado Group) showed the presence of two components of magnetization, carried respectively by magnetite (M) and hematite (H) (see Fig. 2A). While the M component is consistent with those isolated from the other volcanic sites, as shown in Table 1, component H is consistent with those carried by hematite in the sedimentary sites and site F-13 (Table 2). After application of the bedding correction (including plunge of the fold axis) the mean direction statistical parameters deteriorate: K reduces from 14.6 to 9.3 and α_{95} increases from 7.5° to 9.6° (Fig. 4B), suggesting a post-folding magnetization. All analysis were performed on a sample-basis (Van der Voo 1990) due to the fact that most sites are sedimentary and the magnetization resides in the cement. Therefore, a large proportion of secular variation must have taken averaged in each sample. Anyway, no significant directional difference was observed between the overall mean direction reported on a site basis or on a sample basis. The application of McFadden's (1990) fold test indicates a post-folding magnetization. An average of the *in situ* sample remanence directions yields a mean direction at Dec = 148.7° Inc = 25.0° $n = 27$ (samples) $\alpha_{95} = 7.5^\circ$ (Fig. 4B). Dávila *et al.* (2003) have demonstrated a protracted tectonic activity that affected the Famatina region from the Ordovician to the Late Tertiary. A fully post-tectonic magnetization should represent then a Neogene (younger than 10 Ma) average geomagnetic field direction. This is unlikely in the case of component H, since its direction is far away from any Neogene expected direction from the region, even considering potential undetected tectonic rotations. Furthermore, a recent palaeomagnetic study on the middle Miocene Del Crestón Formation, exposed in the area (Zambrano *et al.* in preparation) showed the lack of any significant tectonic rotation since the Miocene. Therefore, a magnetization acquired at sometime between the first and the last tectonic event is inferred. The exclusive reverse polarity of component H and the existence of important successions of red beds of Permian age in the area suggest that a Permian (Kiaman) remagnetization is a possible explanation. If a partial tectonic correction of the structure corresponding to the post-Permian folding and tilting (Dávila *et al.* 2003) is applied, the mean sample direction of component H results in Dec = 171.4°, Inc = 52.6°, $n = 27$, $\alpha_{95} = 7.5^\circ$, $K = 14.6$. This direction resembles the expected Permian direction for this region, according to the South American apparent polar wander path (Rapalini *et al.* 2006) Therefore, a palaeomagnetic pole was calculated using a VGP for each sample after correction by post-Permian tectonic structure, MCM2: 78.7°S 330.8°E, $A_{95} = 7.2^\circ$, $K = 16.1$, $n = 27$.

3.3 Volcanic dyke

A 2.5 m thick igneous dyke (of possible Silurian-Devonian age, Turner 1964) was sampled at site F-20. The subvertical dyke intrudes the Molles Formation with a N–S direction. According to the IRM acquisition curve (Fig. 3A) and the demagnetization diagrams a low-coercivity mineral (MD? magnetite) is the principal magnetic carrier. Well-defined components with MAD between 2° and 7.6° were determined but they do not show any consistency within the site, so no further analysis was done.

3.4 Permian sedimentary rocks

AF and thermal demagnetization (Fig. 2C) and IRM acquisition curves (Fig. 3C) show that hematite is the principal magnetic carrier in these rocks. The initial bulk susceptibility ranges between 4.80×10^{-5} and 1.95×10^{-4} SI. One sample per site was demagnetized with AF techniques with poor results, so 5–6 samples per site were demagnetized by thermal techniques. In all samples (except for P6-4b) a high-temperature magnetic component was isolated (Tables 3 and 4; Fig. 2C).

Samples were collected at two different localities (see Fig. 1): sites P1–P4 were located in the Los Colorados Anticline, while P5–P10 are situated in the Los Damascos thrust sheet. Each locality presented well-grouped site mean as well as sample directions. However, mean directions both *in situ* or after bedding correction differ significantly from one locality to the other (Figs 4C and D). Sites P1–P4 are characterized by northward declinations and negative inclinations, while P5–P10 show southward declinations with positive inclinations. Antipodal directions at both localities are ruled out by a significant failure of the reversal test (McFadden & McElhinny 1990), both *in situ* and after bedding correction. Difference in inclination for both groups amounts to nearly 30°.

Application of the structural corrections does not produce any significant change in the statistical parameters of each group of sites, yielding an indeterminate result for the fold test. This is due to the nearly homoclinal attitude of the sampled beds at each locality. The different directions between both localities suggest a significant different age of magnetization. Lack of resemblance of the *in situ* directions to any post-Permian expected direction suggests a pre-tectonic remanence. To check any possible distortion of the magnetic

Table 1. Sample and site mean characteristic remanence magnetization for the volcanic units of the Molles Formation and Cerro Morado Group (component M).

Site: Unit Lithology	Specimen	Dec (°)	Inc (°)	MAD/α_{95} (°)	Bedding		Dec (°)*	Inc (°)*
					Strike (°)	Dip (°)		
F01: CoM Rhyolites	F1-1a	163.4	41.0	2.9	311	50	98.6	47.9
	F1-2a	167.2	44.6	3.0	311	50	94.3	51.5
	F1-4a	171.1	44.4	2.8	311	50	95.6	54.2
	F1-5a	158.2	43.3	3.0	311	50	93.7	44.9
	F1-7a	160.9	38.9	3.8	311	50	100.5	45.3
	F1-7b	151.9	39.2	1.4	311	50	96.5	39.0
Mean Site F01 Dec = 162.0° Inc = 42.1° α_{95} = 4.7° K = 206.5 n = 6 Dec* = 96.9° Inc* = 46.3° VGP: Lat = 18.4°S Long = 0.2°E								
F02: CoM Ignimbrites	F2-1a	160.8	31.2	0.6	311	50	110.1	42.0
	F2-1b	156.4	37.5	0.7	311	50	100.4	41.5
	F2-2	164.4	55.2	0.7	311	50	77.2	51.0
	F2-6	143.4	29.4	1.7	311	50	102.8	28.2
Mean Site F02 Dec = 155.4° Inc = 38.5° α_{95} = 15.8° K = 34.7 n = 4 Dec* = 99.0° Inc* = 40.4° VGP: Lat = 18.3°S Long = 5.4°E								
F03: CoM Ignimbrites	F3-2a	168.7	39.8	1.0	311	50	102.2	51.3
	F3-3b	169.6	41.2	2.1	311	50	100.4	52.4
	F3-4a	157.0	21.5	1.5	311	50	118.4	34.1
	F3-5a	164.8	38.7	1.6	311	50	102.3	48.1
	F3-7a	157.5	40.2	1.8	311	50	97.5	43.4
	F3-7b	169.0	43.3	0.6	311	50	96.9	52.5
Mean Site F03 Dec = 164.1° Inc = 37.6° α_{95} = 7.7° K = 77.4 n = 6 Dec* = 103.8° Inc* = 46.4° VGP: Lat = 24.1°S Long = 2.6°E								
F04: CoM Ignimbrites	F4-1a	149.2	39.3	4.2	300	59	81.6	41.5
	F4-2a	149.6	38.2	1.1	300	59	83.1	41.6
	F4-3a	159.8	38.0	1.6	300	59	85.0	49.6
	F4-4b	155.3	37.1	1.8	300	59	85.7	45.9
	F4-7a	160.0	36.9	2.9	300	59	86.7	49.6
	F4-7b	157.1	29.8	1.0	300	59	96.4	45.8
Mean Site F04 Dec = 155.2° Inc = 36.6° α_{95} = 4.2° K = 250.4 n = 6 Dec* = 86.5° Inc* = 44.6° VGP: Lat = 9.5°S Long = 357.4°E								
F05: CoM Ignimbrites	F5-1a	162.1	32.0	1.3	300	59	94.7	50.5
	F5-3a	171.1	29.4	0.9	300	59	101.9	57.5
	F5-5a	163.8	33.6	1.9	300	59	92.6	52.2
	F5-5b	173.7	33.4	0.6	300	59	95.0	60.3
	F5-6a	140.2	36.1	2.3	300	59	83.4	33.9
	F5-7b	163.0	27.8	1.0	300	59	101.4	50.2
Mean Site F05 Dec = 162.6° Inc = 32.5° α_{95} = 8.5° K = 62.5 n = 6 Dec* = 94.0° Inc* = 50.0° VGP: Lat = 17.4°S Long = 356.1°E								
F06: CoM Ignimbrites Rhyolites	F6-1b	104.0	54.3	0.8	300	59	54.5	17.1
	F6-3a	153.9	49.2	3.9	300	59	68.2	45.6
	F6-5a	163.3	37.4	1.9	300	59	86.4	52.3
	F6-6a	163.5	39.5	2.1	300	59	83.0	52.5
	F6-7a	123.6	27.1	3.4	300	59	56.3	28.4
Mean Site F06 Dec = 145.3° Inc = 49.9° α_{95} = 18.6° K = 17.9 n = 5 Dec* = 67.7° Inc* = 39.0° VGP: Lat = 7.4°N Long = 352.1°E								
F07: CoM Rhyolites	F7-1a	158.1	29.9	0.5	300	59	96.6	46.7
	F7-1b	153.4	32.1	1.6	300	59	92.1	43.3
	F7-2a	157.0	31.0	0.7	300	59	94.7	46.0
	F7-3a	164.2	23.5	1.9	300	59	108.3	49.8
	F7-4b	158.3	27.4	1.8	300	59	100.1	46.1
	F7-6a	149.3	35.8	1.6	300	59	86.1	40.9
Mean Site F07 Dec = 156.9° Inc = 30.0° α_{95} = 5.0° K = 180.3 n = 6 Dec* = 96.0° Inc* = 44.6° VGP: Lat = 17.1°S Long = 1.2°E								
F08: CoM Rhyolites	F8-1a	166.7	42.7	2.0	300	59	77.5	54.9
	F8-1b	173.9	32.4	0.4	300	59	97.0	60.4
	F8-2a	170.5	38.7	3.3	300	59	84.4	58.0
	F8-3a	168.1	35.2	2.1	300	59	90.7	55.9

© 2008 The Authors, *GJI*, **173,** 63–78
Journal compilation © 2008 RAS

Table 1. (*Continued.*)

Site: Unit Lithology	Specimen	Dec (°)	Inc (°)	MAD/α_{95} (°)	Bedding		Dec (°)*	Inc (°)*
					Strike (°)	Dip (°)		
	F8-6a	168.3	31.9	1.2	300	59	96.6	55.6
	F8-7a	165.3	34.9	1.4	300	59	90.8	53.6
	Mean Site F08 Dec = 168.8° Inc = 36.0° α_{95} = 4.0° K = 286.6 n = 6 Dec* = 89.4° Inc* = 55.5°							
	VGP: Lat = 16°S Long = 349.5°E							
F09: CoM Rhyolites	F9-2a	186.5	45.0	1.3	329	60	95.5	46.7
	F9-3b	190.1	36.9	0.6	329	60	107.7	49.4
	F9-4a	189.5	27.5	0.5	329	60	121.5	47.3
	F9-4b	186.5	32.5	0.9	329	60	113.6	45.9
	F9-5a	189.5	36.1	2.2	329	60	108.9	48.8
	F9-6b	190.8	35.2	1.1	329	60	110.4	49.8
	Mean Site F09 Dec = 188.9° Inc = 35.5° α_{95} = 4.9° K = 186.1 n = 6 Dec* = 110.8° Inc* = 47.9°							
	VGP: Lat = 30.3°S Long = 3.6°E							
F10: CoM Rhyolites	F10-2b	202.0	41.6	3.4	329	60	98.9	58.3
	F10-4a	199.8	29.1	1.0	329	60	122.2	56.5
	F10-5b	197.0	31.0	1.4	329	60	118.2	54.4
	F10-6b	207.7	36.0	1.9	329	60	109.1	63.5
	F10-7a	203.0	47.5	0.7	329	60	87.9	57.4
	F10-7b	198.7	35.4	2.0	329	60	110.7	56.2
	Mean Site F10 Dec = 201.2° Inc = 36.8° α_{95} = 6.2° K = 117.4 n = 6 Dec* = 109.5° Inc* = 57.9°							
	VGP: Lat = 31.9°S Long = 352.4°E							
F11: CoM Rhyolites	F11-1a	172.6	42.4	2.7	329	60	98.2	36.6
	F11-2a	174.8	36.0	2.6	329	60	106.5	37.1
	F11-4a	185.1	50.5	1.1	329	60	87.7	45.3
	F11-4b	175.4	45.8	2.2	329	60	94.3	39.0
	F11-5a	179.3	39.6	2.6	329	60	102.7	41.2
	F11-6b	178.9	40.8	0.8	329	60	101.1	41.0
	Mean Site F11 Dec = 177.5° Inc = 42.6° α_{95} = 4.9° K = 184.5 n = 6 Dec* = 99.6° Inc* = 39.9°							
	VGP: Lat = 18.7°S Long = 6.0°E							
F12: CoM Rhyolites	F12-1b	182.1	48.9	2.7	329	60	90.2	43.5
	F12-2a	176.0	57.0	0.7	329	60	79.8	38.3
	F12-2b	175.6	54.0	1.3	329	60	83.7	39.2
	F12-4b	178.6	55.4	3.7	329	60	81.7	40.9
	F12-5a	177.3	55.6	2.3	329	60	81.5	40.1
	F12-6a	178.2	55.4	3.5	329	60	81.7	40.6
	Mean Site F12 Dec = 178.1° Inc = 54.4° α_{95} = 2.6° K = 649.6 n = 6 Dec* = 84.0° Inc* = 40.6°							
	VGP: Lat = 6.0°S Long = 359.0°E							
F16: CoM Rhyolites	F16-1a	109.1	26.5	2.3	219	52	78.8	69.0
	F16-2	106.6	31.5	6.8	219	52	63.4	70.0
	F16-3a	124.4	10.9	3.4	219	52	128.5	61.7
	F16-4a	121.9	15.9	5.7	219	52	120.8	66.1
	F16-4b	110.1	24.2	1.7	219	52	85.1	68.1
	F16-6a	106.3	25.2	4.0	219	52	77.1	66.2
	Mean Site F16 Dec = 113.4° Inc = 22.5° α_{95} = 8.8° K = 58.5 n = 6 Dec* = 93.6° Inc* = 69.5°							
	VGP: Lat = 24.7°S Long = 333.3°E							
F17: Mo Basaltic flow	F17-3a	88.0	1.4	1.3	205	50	87.8	43.3
	F17-3b	77.9	–3.8	2.3	205	50	80.2	33.7
	F17-4a	90.0	0.8	3.5	205	50	90.6	43.7
	F17-5b	82.6	–6.6	4.4	205	50	86.7	33.7
	F17-6c	75.7	–2.0	2.9	205	50	76.8	33.9
	F17-7b	84.7	–3.9	5.1	205	50	87.4	37.1
	Mean Site F17 Dec = 83.2° Inc = –2.4° α_{95} = 5.3° K = 162.1 n = 6 Dec* = 84.6° Inc* = 38.6°							
	VGP: Lat = 5.8°S Long = 0.5°E							
Samples mean	Dec = 157.0° Inc = 39.5° α_{95} = 6.0° K = 7.8 R = 71.67 n = 82							
	Dec* = 92.5° Inc* = 48.0° α_{95}* = 3.0° K* = 27.57 R = 79.06							
Sites mean	Dec = 157.2° Inc = 38.8° α_{95} = 14.3° K = 8.7 R = 12.5 N = 14							
	Dec* = 93.4° Inc* = 47.7° α_{95}* = 5.8° $K2$* = 48.5 R = 13.7							
	Mean VGPs: Lat: 16.7°S Long: 357.2°E A_{95} = 6.5° K = 38.5 N = 14							

Notes: Bedding correction includes strike and dip of the units and 'unplunging' of the fold axis (350°/33°). CoM: Cerro Morado Group; Mo: Molles Formation. K and α_{95}: statistical parameters of Fisher (1953). Dec*, Inc*: declination and inclination after structural correction. Lat: latitude, Long: longitude.

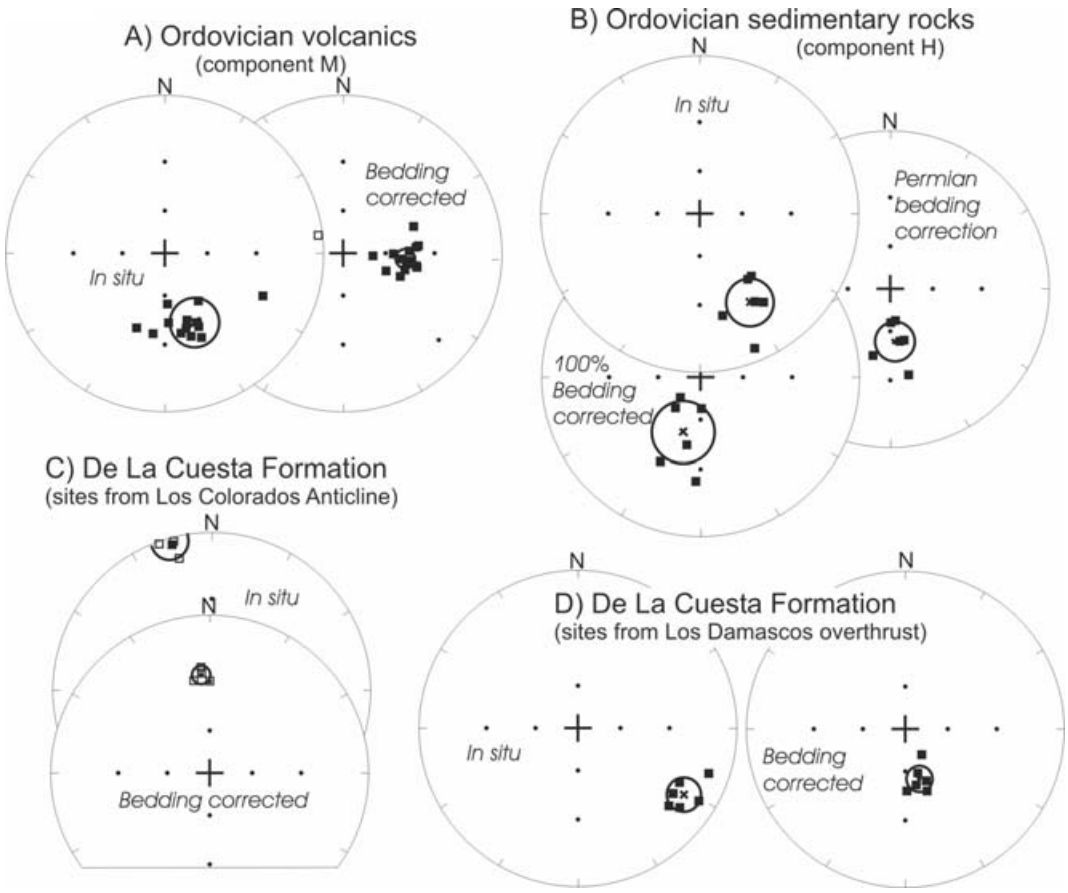


Figure 4. (A) *In situ* and bedding corrected distribution of site mean directions for the Ordovician volcanics (component M). (B) Sample remanence directions of component H (mainly Ordovician sedimentary rocks), *in situ*, after 100 per cent bedding correction and after partial structural correction for the attitude of the Permian redbeds. (C) Sample characteristic remanence directions for the Permian sedimentary rocks of the De La Cuesta Formation sampled at the Los Colorados Anticline: *in situ* and after 100 per cent bedding correction. (D) *In situ* and bedding corrected sample characteristic remanence directions of the Permian sedimentary rocks of the De La Cuesta Formation exposed at the Los Damascos overthrust. Equal-angle stereoplot, open (closed) symbol corresponds to negative (positive) inclination.

field recorded by the rocks due to compaction induced flattening of the remanence, an inclination test was performed (Hodych & Buchan 1994; Rapalini 2006). After measuring the NRM, increasing magnetic fields (150, 250, 350, 450, 600, 800 and 1000 mT) were applied with a pulse magnetizer, at nearly 45° respect to the bedding plane of each sample (1 or 2 samples per site were used). Any significant systematic deflection of the IRM towards the bedding plane would suggest that a correction of the characteristic remanence directions is needed due to inclination shallowing. Fig. 5 shows that in all cases the IRM inclination measured after each step is virtually indistinguishable from that of the applied field. This rules out any significant inclination shallowing and therefore the characteristic mean remanence direction for each site can be considered as an accurate record of the mean palaeomagnetic field.

A sample-based mean (Van der Voo 1990) of the bedding corrected remanence directions for P1–P4 yields an average direction at Dec = 355.4°, Inc = –26.1°, n(number of samples) = 21, $\alpha_{95} = 2.4°$, $K = 190.7$; and for P5–P10, Dec = 168.0°, Inc = 54.1°, $n = 29$, $\alpha_{95} = 4.7°$, $K = 34.0$. By averaging the VGP, of each sample, one palaeomagnetic pole was calculated for each group P1–P4 (LC2): 74.5°N, 275.4°E, $A_{95} = 2.0°$, $K = 258.3$ and P5–P10 (LC1): 76.9°S, 345.2°E, $A_{95} = 9.4°$, $K = 21.1$.

4 INTERPRETATION AND DISCUSSIONS

The Early Ordovician pole obtained from the volcanic rocks of the Cerro Morado Group and Molles Formation (MCM1, Fig. 6) does not coincide with the reference pole of Gondwana for 475 Ma (Grunow 1995). This anomaly is due to a clockwise rotation of 38.5° ± 7.8° around a vertical axis with an anomaly in inclination scarcely significant (–8.8° ± 7.0°). The age of this rotation can be constrained by analysing the other palaeomagnetic results obtained in this study.

The MCM2 and LC1 poles are virtually identical (Fig. 7), including the exclusive reverse polarity consistent with a Kiaman magnetization age. The position on the South American APWP (Fig. 7) suggests an age of magnetization near 270 Ma in the early Late Permian and close to the end of the Kiaman superchron (Gradstein *et al.* 2004). The simplest interpretation indicates a primary (early diagenetic?) magnetization of the Permian De La Cuesta Formation (LC1) and a remagnetization of the Early Ordovician clastic sedimentary rocks of the Molles Formation, possibly associated to a regional diagenetic event that produced precipitation of hematite, perhaps through pervasive percolation of fluids associated with weathering

© 2008 The Authors, *GJI*, **173,** 63–78
Journal compilation © 2008 RAS

Table 2. Sample and site mean characteristic remanence magnetization for the sedimentary units of Molles Formation and Cerro Morado Group (component H).

					Bedding						VGP	
Site: Unit Lithology	Sample	D (°)	I (°)	MAD/α_{95} (°)	Strike (°)	Dip (°)	D (°) (*)	I (°) (*)	D (°) (**)	I (°) (**)	Lat	Long
F13: Mo Rhyolites	13-1a	171.7	5.1	6.3	187	47	193.1	14.8	183.1	24.2	–73.7	123.0
	13-2a	140.9	19.5	8.0	187	47	180.7	46.5	157.1	50.8	–70.0	8.4
	13-5	146.5	31.3	3.9	187	47	199.9	49.4	174.4	58.9	–78.1	313.6
	13-7a	140.3	27.9	3.8	187	47	190.1	51.6	162.1	58.7	–71.8	341.5
	13-10b	137.5	30.7	5.0	187	47	191.7	55.4	160.1	62.3	–68.2	333.8
	Mean Site F13	147.8	23.5	16.5	$K = 22.5$	$n = 5$	191.2	43.9	169.1	51.6	–78.5	352.4
F14: Mo Sandstones	14-3a	164.6	20.5	2.9	187	47	200.5	29.8	186.0	40.9	–82.5	159.1
	14-5a	171.4	26.4	2.4	187	47	210.4	28.4	197.4	41.9	–73.8	190.0
	Mean Site F14	167.9	23.5	18.8	$K = 178.3$	$n = 2$	205.5	29.2	191.7	41.5	–78.5	180.3
F15: Mo Sandstones	15-1a	138.9	–32.1	5.8	187	47	144.5	6.0	138.6	1.5	–41.6	50.0
	15-2a	151.5	10.7	2.6	187	47	182.0	32.6	165.4	38.6	–75.1	46.6
	15-3a	153.4	40.2	3.9	187	47	216.4	48.5	194.9	62.1	–71.1	257.0
	15-4a	179.5	–8.7	1.9	187	47	189.2	0.1	182.5	8.4	–75.2	285.1
	15-6a	162.8	12.6	6.2	187	47	192.2	26.1	178.8	35.1	–80.6	105.3
	Mean Site F15	157.9	4.7	30.6	$K = 7.21$	$n = 5$	182.7	24.3	169.3	30.5	–81.2	30.3
F16: CoM Rhyolites	16-1a	160.3	12.9	8.0	219	52	219.2	77.6	176.4	36.6	–81.1	90.0
	16-2	144.2	35.7	1.4	219	52	188.2	57.3	176.4	63.7	–73.1	300.9
	16-3a	154.1	14.0	2.2	219	52	188.2	57.3	170.3	40.5	–79.7	52.6
	16-4b	112.1	17.5	4.1	219	52	98.8	64.2	112.2	52.9	–32.8	358.9
	16-6a	146.3	21.3	10.8	219	52	184.3	67.5	165.5	50.3	–77.2	8.0
	Mean Site F16	143.7	21.1	19.2	$K = 16.8$	$n = 5$	178.6	68.5	162.0	51.2	–73.5	2.7
F18: Mo Sandstones	18-1	145.8	31.7	12.3	200	53	211.4	60.5	173.9	59.6	–77.2	313.7
	18-3b	143.1	37.5	10.1	200	53	222.7	63.8	177.3	65.6	–70.8	297.7
	18-5a	164.5	37.1	16.9	200	53	230.0	47.5	201.8	53.7	–70.6	224.3
	18-6a	133.3	38.3	5	200	53	222.3	71.8	161.1	70.6	–60.5	314.5
	18-7a	136.2	34.4	10.8	200	53	212.1	68.6	161.7	66.1	–65.7	322.7
	Mean Site F18	144.5	36.6	9.7	$K = 63.7$	$n = 5$	220.7	62.7	177.6	64.0	–72.3	300.0
F19: Mo Sandstones Limestones	19-3a	132.3	43.5	5.2	210	51	253.4	73.5	168.6	75.4	–55.4	301.5
	19-4a	139.1	29.9	5.3	210	51	209.7	67.9	162.0	60.9	–70.3	335.2
	19-5a	134.6	39	4.5	210	51	238.5	73.2	164.8	70.7	–61.5	310.6
	19-6b	144.6	40.7	5.0	210	51	230.1	71.2	184.7	67.3	–68.3	284.0
	19-7a	149.6	28.2	8.3	210	51	214.7	59.3	175.4	54.9	–82.2	321.1
	Mean Site F19	140.3	36.4	8.4	$K = 83.1$	$n = 5$	226.4	69.7	171.4	66.1	–68.4	307.7
Samples mean	Dec = 148.7° Inc = 25.0° $\alpha_{95} = 7.5°$ $K = 14.6$ $R = 25.2$ $n = 27$ Dec* = 197.3° Inc* = 53.5° $\alpha_{95}{}^{*} = 9.6°$ $K^{*} = 9.3$ $R^{*} = 24.2$ Dec** = 171.4° Inc** = 52.6° $\alpha_{95}{}^{**} = 7.5°$ $K^{**} = 14.6$ $R^{**} = 25.2$											
	Mean VGPs (samples): Lat: 78.7°S Long: 330.8°E $A_{95} = 7.2°$ $K = 16.1$ $n = 27$											
Sites mean	Dec = 152.1° Inc = 24.6° $\alpha_{95} = 12.3°$ $K = 30.4$ $R = 5.8$ $N = 6$ Dec* = 199.1° Inc* = 50.3° $\alpha_{95}{}^{*} = 18.9°$ $K^{*} = 13.5$ $R = 5.6$ Dec** = 173.7° Inc** = 51.3 $\alpha_{95}{}^{**} = 12.6$ $K^{**} = 29.0$ $R^{**} = 5.8$											
	Mean VGPs (sites): Lat: 80.9°S Long: 328.2°E $A_{95} = 11.2°$ $K = 36.7$ $N = 6$											

(*) Bedding correction includes strike and dip of the units and 'unplunging' of the fold axis (350°/33°). (**) Permian bedding correction strike: 207°, dipping: 36° and 'unpluging' of the Permian fold axis (16°/10°). K and α_{95}: statistical parameters of Fisher (1953). Lat: latitude; negative correspond to southern latitude, Long: longitude.

and secondary leaching. The fact that most Ordovician volcanic rocks were not affected by such signal suggests that rock porosity may have played a selective role for remagnetization. Similar mechanisms have been interpreted for Australian (Geeve *et al.* 2001) and Appalachian (McCabe *et al.* 1989) rocks. A common origin for hematite in the De La Cuesta and Molles Formations is supported by identical magnetic characteristics as well as similar mineralogical textures observed in thin and polished sections. The presence of one volcanic site of the Cerro Morado Group (F-16) with both the primary magnetization carried by magnetite (component M) and the secondary one by hematite (component H) strongly suggests a chemical origin for the latter. The consistency of LC1 and MCM2 poles with the South American apparent polar path indicates lack of significant tectonic rotations in the area since the Permian and constrains the rotation of the Early Ordovician volcanics to pre-Permian times.

Sites P1–P4 sampled at the Los Colorados Anticline provided a second palaeomagnetic pole (LC2) for the De La Cuesta Formation. Although no fossil assemblages have been described for these outcrops that could determine their age beyond reasonable

Table 3. Sample and site mean characteristic remanence magnetization for the De La Cuesta Formation (sites P1–P4).

					Bedding				VGP	
Site	Sample	Dec (°)	Inc (°)	MAD/α_{95} (°)	Strike (°)	Dip (°)	Dec (°) (*)	Inc (°) (*)	Lat	Long
P01	P1-3a	336.7	6.0	3.8	210	37	344.6	–23.6	68.2°N	247.8°E
	P1-4b	335.3	–4.1	5.6	210	37	348.1	–33.1	74.8°N	243.9°E
	P1-5b	344.6	–4.2	7.6	210	37	357.5	–29.1	76.7°N	281.7°E
	P1-6a	341.0	0.9	3.9	210	37	352.4	–27.0	74.0°N	264.6°E
	Mean Site P01	339.4	–0.4	7.3	$K = 158.6$	$n = 4$	350.6	–28.3	73.9°N	257.7°E
P02	P2-1b	340.6	–1.0	5.7	210	37	351.8	–28.3	74.4°N	261.4°E
	P2-2a	341.5	3.8	3.2	210	37	350.9	–24.3	71.9°N	262.4°E
	P2-3a	349.3	3.0	4.1	210	37	358.6	–20.6	71.9°N	287.8°E
	P2-4a	345.1	2.7	3.3	210	37	355.3	–23.4	72.9°N	276.4°E
	P2-5b	344.1	2.3	6.8	210	37	354.3	–23.8	72.9°N	272.9°E
	Mean Site P02	344.1	2.2	3.7	$K = 433.19$	$n = 5$	354.2	–24.1	73.0°N	272.4°E
P03	P3-1b	342.8	–0.4	8.9	207	35	351.9	–23.1	71.7°N	266.2°E
	P3-2b	343.1	–4.1	5.4	207	35	354.9	–26.2	74.4°N	273.5°E
	P3-4b	349.9	–1.8	8.3	207	35	359.1	–20.8	72.0°N	289.3°E
	P3-5b	348.1	3.4	4.6	207	35	356.2	–17.7	70.0°N	281.1°E
	P3-6a	343.0	–4.5	5.0	207	35	354.9	–26.2	74.4°N	273.5°E
	Mean Site P03	345.4	–1.5	4.4	$K = 304.3$	$n = 5$	355.4	–22.8	72.7°N	276.9°E
P04	P4-1a	341.8	–11.6	5.8	205	36	358.1	–33.6	79.5°N	282.2°E
	P4-1b	346.3	–8.7	10.0	205	36	1.2	–28.5	76.4°N	297.1°E
	P4.2a	347.3	–6	4.8	205	36	1.0	–26.3	75.2°N	296.0°E
	P4-3b	347.6	–2.7	10.0	205	36	358.7	–22.8	73.1°N	287.8°E
	P4-5b	342.2	–8.4	6.6	205	36	357.3	–30.5	77.5°N	280.2°E
	P4-6a	345.6	–11.8	2.7	205	36	2.0	–31.6	78.3°N	301.6°E
	Mean Site P04	345.1	–8.2	3.5	$K = 359.8$	$n = 6$	359.7	–28.9	76.8°N	291.0°E
Samples mean	Dec = 343.6° Inc = –2.3° $\alpha_{95} = 2.5°$ $K = 162.4$ $R = 20.9$ $n = 21$ Dec* = 355.4° Inc* = –26.1° α_{95}* = 2.4° K* = 190.7 R* = 19.9									
	Mean VGPs (samples): Lat: 74.5°N Long: 275.4°E $A_{95} = 2.0°$ $K = 258.3$ $n = 21$									
Sites mean	Dec = 343.4° Inc = –1.8° $\alpha_{95} = 5.8°$ $K = 250.7$ $R = 4.0$ $N = 4$ Dec* = 355.0° Inc* = –26.1° α_{95}* = 5.1° K* = 323.6 R* = 4.0									
	Mean VGPs (sites): Lat: 74.4°N Long: 273.8°E $A_{95} = 4.6°$ $K = 401.9$ $N = 4$									

All samples are corrected by the bedding plane and by the anticline structure (8°/16°). Dec*, Inc*: declination and inclination after structural correction. K and α_{95}: statistical parameters of Fisher (1953). Lat: latitude, Long: longitude.

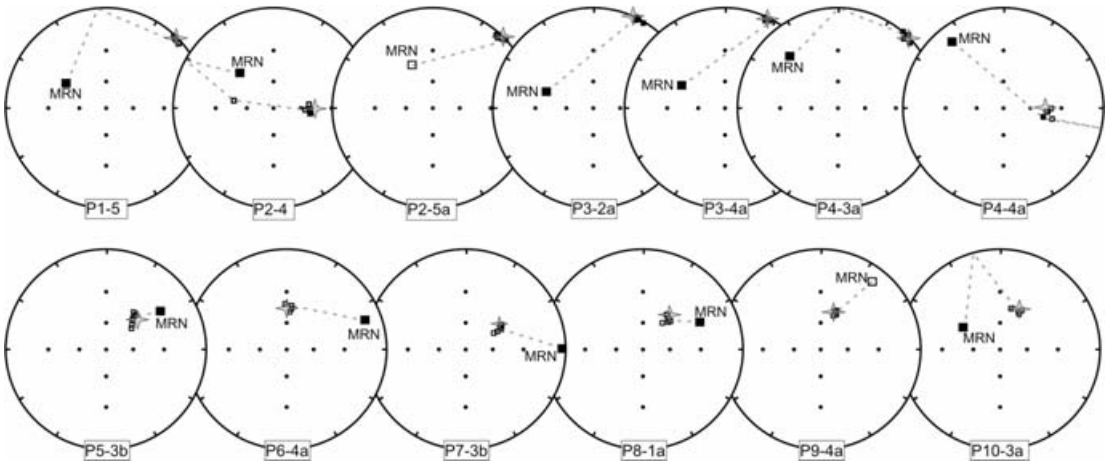


Figure 5. Illustration of the test for inclination shallowing performed on the Permian sedimentary rocks. Equal-angle stereoplots show the directions of the applied magnetic field (grey star), the NRM of the specimen (big square) and the acquired magnetizations, in field coordinates. Note that there is no significant shallowing of inclination.

doubts, according to stratigraphic studies (Dávila 2003) they should be slightly older than those sampled at Los Damascos thrust sheet (P5–P10) that yielded pole LC1. Therefore an exclusive reverse polarity corresponding to the Reversed Kiaman Superchron should be expected. However, only normal polarities were recovered from these samples. The obtained pole is not consistent either with the Permian section of the South American APWP nor can it be reconciled by any rotation around a vertical axis. It is consistent, however,

© 2008 The Authors, *GJI*, **173,** 63–78
Journal compilation © 2008 RAS

Table 4. Sample and site mean characteristic remanence magnetization for the De La Cuesta Formation (sites P5 to P10).

					Bedding				VGP	
Site	Sample	Dec (°)	Inc (°)	MAD/α_{95}(°)	Strike	Dip	Dec (°) (*)	Inc (°) (*)	Lat	Long
P05	P5-1	100.8	2.0	7.2	179	70	122.9	69.1	42.1°S	335.7°E
	P5-2b	103.5	7.5	10.1	179	70	139.5	71.7	50.2°S	326.4°E
	P5-3a	99.9	0.7	4.3	179	70	118.9	68.4	39.5°S	337.3°E
	P5-5b	118.5	6.5	14.3	179	70	158.5	58.8	69.2°S	345.2°E
	P5-6b	122.0	24.7	10.5	179	70	194.9	59.4	73.2°S	249.5°E
	Mean Site P05	108.7	8.3	13.4	$K = 33.4$	$n = 5$	150.7	68.2	56.2°S	329.3°E
P06	P6-1a	126.1	36.4	7.1	180	61	203.0	58.1	68.5°S	236.2°E
	P6-3	124.8	24.4	5.2	180	61	179.8	58.5	79.5°S	293.1°E
	P6-5b	143.1	21.6	9.7	180	61	183.8	41.6	84.1°S	148.7°E
	P6-6a	129.0	24.2	5.8	180	61	181.4	54.8	83.3°S	282.5°E
	Mean Site P06	130.9	26.8	11.5	$K = 64.7$	n = 4	186.4	53.8	81.7°S	249.9°E
P07	P7-1b	127.6	10.5	9.5	174	62	159.6	45.4	71.9°S	23.2°E
	P7-2a	130.2	20.0	3.5	174	62	173.6	47.1	84.4°S	25.0°E
	P7-3a	123.9	22.4	4.5	174	62	174.2	53.4	82.8°S	334.2°E
	P7-4b	106.9	18.4	7.4	174	62	154.4	66.8	61.6°S	328.5°E
	P7-6b	103.2	22.6	10.6	174	62	161.0	71.7	59.1°S	312.8°E
	Mean Site P07	118.5	19.1	12.1	$K = 40.7$	$n = 5$	165.8	57.1	74.0°S	337.4°E
P08	P8-1b	129.3	10.6	8.8	173	62	160.5	43.1	72.3°S	29.5°E
	P8-2a	141.4	18.7	4.7	173	62	176.4	35.9	80.6°S	91.1°E
	P8-3a	131.9	12.3	4.1	173	62	164.2	41.7	75.1°S	37.0°E
	P8-4b	128.3	8.6	8.9	173	62	157.5	43.0	69.6°S	27.5°E
	P8-5A	111.7	11.2	6.4	173	62	146.8	58.2	60.8°S	353.1°E
	Mean Site P08	128.5	12.4	10.6	$K = 53.4$	$n = 5$	162.2	44.7	73.7°S	24.3°E
P09	P9-1a	120.3	23.1	4.3	169	74	186.4	50.8	83.8°S	230.8°E
	P9-2b	124.2	18.0	11.2	169	74	178.5	46.8	88.5°S	48.8°E
	P9-3a	125.2	14.2	1.2	169	74	173.4	45.7	84.0°S	35.8°E
	P9-5a	125.7	19.0	5.8	169	74	180.3	45.8	88.5°S	122.5°E
	P9-6b	126.5	18.6	12.3	169	74	179.5	44.7	87.6°S	101.6°E
	Mean Site P09	124.4	18.6	3.7	$K = 431.54$	$n = 5$	179.6	47.0	89.3°S	79.4°E
P10	P10-1b	123.6	7.0	4.2	173	70	160.8	48.9	73.2°S	13.8°E
	P10-2b	124.2	8.1	8.9	173	70	162.7	48.5	74.9°S	15.2°E
	P10-3b	113.9	11.6	5.2	173	70	162.4	59.5	71.5°S	338.8°E
	P10-4b	126.2	5.5	7.2	173	70	160.2	45.7	72.5°S	22.7°E
	P10-5a	118.2	4.9	5.1	173	70	154.3	52.9	67.5°S	3.6°E
	Mean Site P10	121.2	7.4	5.4	$K = 205.4$	$n = 5$	160.0	51.2	72.5°S	6.2°E
Samples mean	Dec = 121.6° Inc = 15.2° $\alpha_{95} = 4.5°$ $K = 36.7$ R = 28.2 $n = 29$ Dec* = 168.0° Inc* = 54.1° α_{95}* = 4.7° K* = 34.0									
	Mean VGPs (samples): Lat: 76.9°S Long: 345.2°E $A_{95} = 6.0°$ $K = 21.1$ $n = 29$									
Sites mean	Dec = 121.9° Inc = 15.5° $\alpha_{95} = 8.9°$ $K = 57.7$ $N = 6$ Dec* = 164.2° Inc* = 54.2° α_{95}* = 9.4° K* = 52.3									
	Mean VGPs (sites): Lat: 77.2°S Long: 343.6°E $A_{95} = 12.2°$ $K = 31.3$ $N = 6$									

Dec*, Inc*: declination and inclination after structural correction. K and α_{95}: statistical parameters of Fisher (1953). Lat: latitude, Long: longitude.

with Early Cretaceous poles (160–130 Ma) from South America (Fig. 7, Tamrat & Ernesto 2006). Since the IRM inclination experiments described above indicated that no significant inclination shallowing affects these rocks, two possible interpretations are suggested. The first one suggests that the sampled interval has been erroneously assigned to the Permian De La Cuesta Formation and that it corresponds to a yet unrecognized Early Cretaceous succession. Although erroneous age assignment of red beds successions is frequent, and there are several basins with continental red beds of Cretaceous age in Argentina (Ramos 1999), no such succession has been described in the Famatina ranges yet. Careful stratigraphic work and detailed strata correlation in the area (Dávila 2003; Astini *et al.* in press) demonstrates that this red bed succession corresponds to a lacustrine-ephemeral stream interval immediately beneath the eolian upper member of the Permian De la Cuesta Fm. This suggests that an erroneous age assignment, albeit possible, is unlikely. An alternative could be that these rocks acquired the isolated magnetization during the Early Cretaceous. Red beds remagnetization is very common (Turner 1980) so it could have happened to sites P1–P4 during those times. Dávila (2003) has dated uplift and

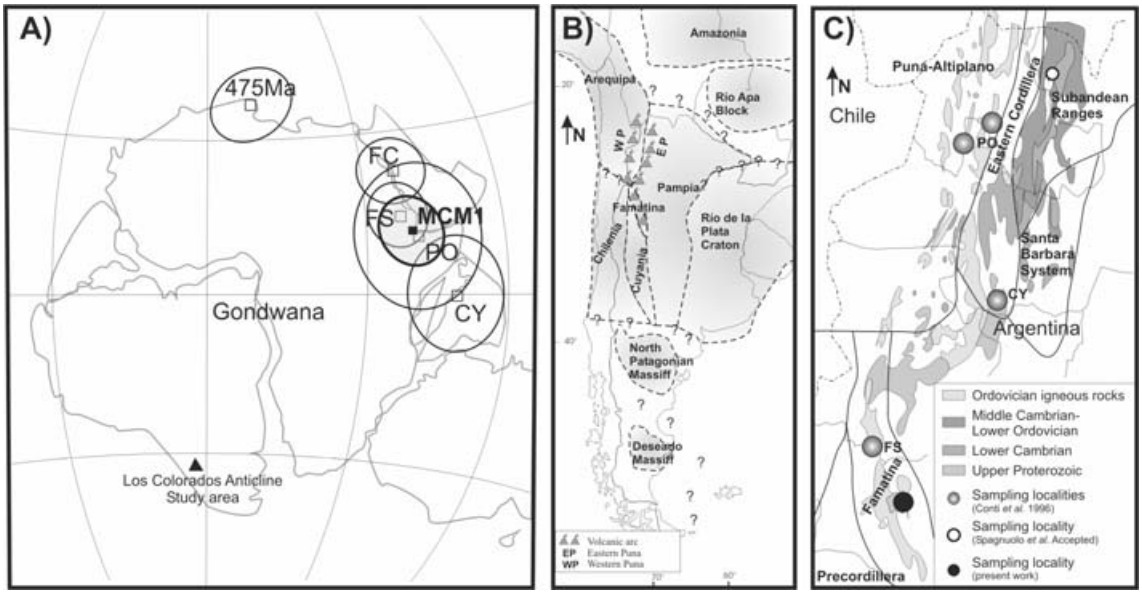


Figure 6. (A) Palaeomagnetic poles from Early Ordovician units of NW Argentina (Conti *et al.* 1996 and this work), the ca. 475 Ma mean pole of Gondwana (Grunow 1995) and the location of the study region in Gondwana. The supercontinent is reconstructed according to Lottes & Rowley (1990) in African coordinates. MCM1: Cerro Morado and Molles volcanics (this work), FS: Famatina System, PO: Eastern Puna, CY: Cuchillaco Granodiorite, FC: Campanario Formation (Spagnuolo *et al.* 2007). (B) Geotectonic map with the main Precambrian and Palaeozoic terranes of South America (modified from Pankhurst *et al*. 2006, Rapalini 2005). (C) Location of the sampling localities of Early Palaeozoic rocks from NW Argentina with significant amounts of clockwise rotations (Conti *et al.* 1996; Spagnuolo *et al.* 2007; this work).

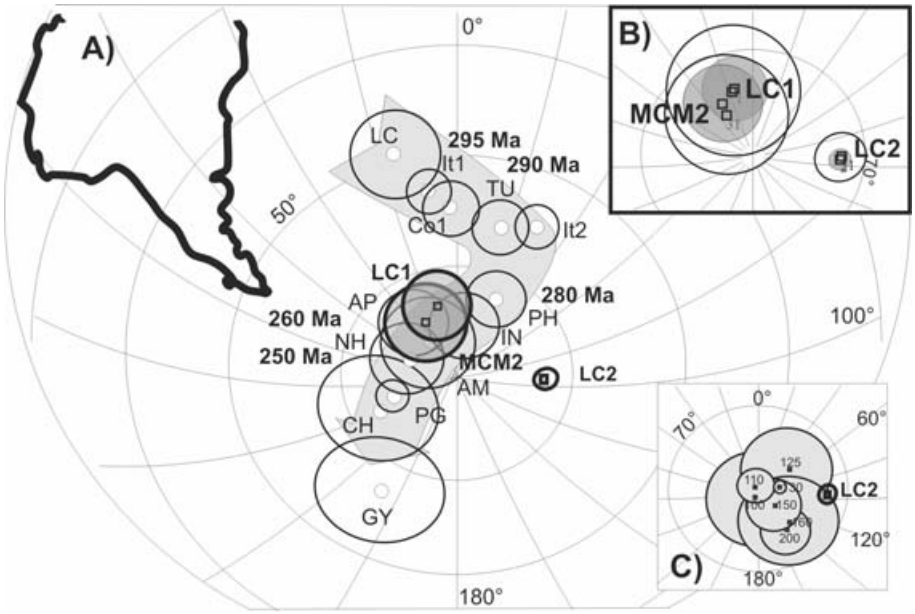


Figure 7. (A) Late Palaeozoic South American palaeomangetic poles (Rapalini *et al.* 2006 and references therein), numbers indicate the more likely age of each pole in m.y. MCM2: pole from component H of the Molles Formation and the Cerro Morado Group (mainly sedimentary units), LC1: De La Cuesta Formation (Los Damascos outcrop). (B) Compared VGP's: in grey the samples-based mean and in white the sites-based mean. There is no difference in both directions of each pole. (C) LC2 (pole from De La Cuesta Formation, Los Colorados Anticline outcrop) and Cretaceous palaeomangetic poles of South America (Besse & Courtillot 2002; Tamrat & Ernesto 2006).

denudation of the Famatina range in the study area as Cretaceous, compatible with similar data obtained by Coughlin *et al.* (1998) in Valle Fértil, Maz, Umango, Famatina and Aconquija Ranges which could be related to a chemical or thermoviscous remagnetization of these rocks.

The most significant result obtained in our study is undoubtedly the Early Ordovician MCM1 pole. As already mentioned, this pole is anomalous respect to the coeval reference pole for Gondwana indicating a significant clockwise rotation of around 39° with virtually no anomaly in palaeolatitude. MCM1 is, however, consistent with coeval poles from the Famatina Range (Valencio *et al.* 1980; Conti *et al.* 1996), the Eastern Puna magmatic belt (Conti *et al.* 1996) and the Cuchiyaco Granodiorite in NW Argentina (Conti *et al.* 1996). All these poles show a significant (30°–55°)

© 2008 The Authors, *GJI*, **173,** 63–78
Journal compilation © 2008 RAS

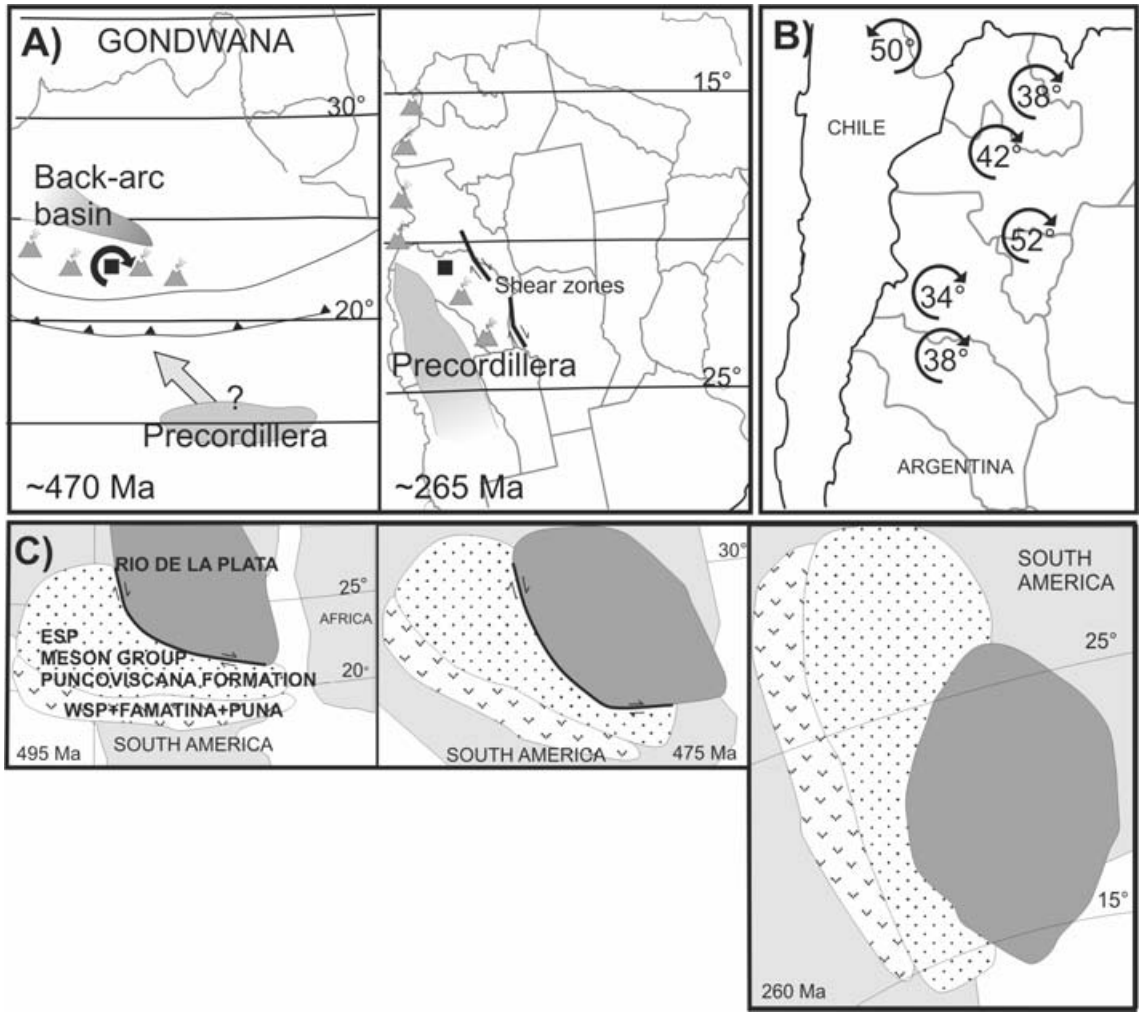


Figure 8. Schematic cartoons of the alternatives presented in the text. (A) The rotation of the Famatina arc is related to the closure of the backarc basin because of the collision of the Precordillera terrane(∼450 Ma). The Precordillera also rotated during this event (Rapalini & Cingolani 2004). The black square is the location of the study area. (B) The systematic block rotations because of oblique subductuion or scape tectonics related to the collision of a terrane. (C) The Famatina area and the northwestern Argentina exerted dextral movements respect to the Rio de la Plata craton during the final stages of amalgamation of Gondwana (Rapela *et al*. 2007). ESP: Eastern Sierras Pampeanas. WSP: Western Sierras Pampeanas.

clockwise rotation of the sampling localities with no palaeolatitude anomaly. Based on those previous results Conti *et al.* (1996) proposed that the Famatina and the Eastern Puna constituted a single para-authocthonous terrane upon which a magmatic arc developed in the Early Ordovician. According to this hypothesis this terrane rigidly rotated clockwise against the SW Gondwana margin, presumably in the Middle or Late Ordovician, possibly associated to the accretion of the allochthonous Precordillera (or Cuyania) terrane in the Late Ordovician (e.g. Thomas & Astini 2003; Ramos 2004). Palaeomagnetic weaknesses of this model were the small number of sites and samples at each locality and the lack of control for the age of rotation. Our new results confirm with a much larger database the previous palaeomagnetic results of Conti *et al.* (1996). As already discussed, the remanence direction obtained from the Permian De la Cuesta Formation and the remagnetization found in the Ordovician sediments, both consistent with the expected Permian direction, allow constraining the age of rotation as pre-Permian. Our study locality is situated some 100 km south from that studied by Valencio *et al.* (1980) and Conti *et al.* (1996), extending the region with such palaeomagnetic signature significantly to the south. As already mentioned, this model is compatible with geophysical, biogeographic and magmatic evidence. The simplest way for a terrane several hundred kilometres long as the Famatina-Eastern Puna, to rotate, perhaps associated to initial stages of Precordillera docking, is by closing a V-shaped basin on its East margin (Fig. 8A). The clockwise sense of rotation prescribes a basin widening towards the North (present-day coordinates). The amount of rotation also indicates that compression of a stretched continental lithosphere would likely be insufficient to accommodate it, and that an oceanic backarc basin should have developed to the East of the Famatina and Eastern Puna. Miller & Sollner (2005) have proposed the existence of an ocean-floored backarc basin to the East of Famatina in the Early Palaeozoic, just what is needed by the rotated terrane hypothesis. However, if the Eastern Puna and Famatina were part of a single crustal block, a much wider basin must have existed to the East of the Eastern Puna magmatic belt. No evidence of such a basin in Late Cambrian to Early Ordovician times has been reported so far, placing a significant difficulty for the geological viability of the model by Conti *et al*. (1996). As a conclusion, the palaeomagnetic and geological evidence is consistent with such model only for Famatina

but seems inadequate for the Eastern Puna, despite the consistent palaeomagnetic results in both regions. More palaeomagnetic data may be needed to relate the kinematics of those areas.

An alternative interpretation is to consider a pattern of systematic clockwise rotations affecting large areas of the SW continental margin of Gondwana in the Early Palaeozoic (Fig. 8B). Different mechanisms could have caused such systematic pattern, from highly oblique subduction to escape tectonics associated with collision of the Precordillera or Chilenia terranes (Ramos *et al.* 1986). Whether any of these mechanisms is likely to have caused these rotations must be tested against available stratigraphic and structural evidence as well as new palaeomagnetic data.

The consistently anomalous pole positions obtained from early Ordovician units of the Famatina and the Eastern Puna magmatic belts strongly advocates homogeneous kinematic behaviour and crustal continuity. Spagnuolo *et al.* (2007) have recently obtained the first Middle to Late Cambrian palaeomagnetic pole for the Palaeozoic basin of Northwest Argentina (Campanario Fm). This basin developed to the East of the Eastern Puna belt on basement that has been assigned to the Pampia block (see Rapalini 2005, and references therein). The Campanario Fm pole is also anomalous respect to the Late Cambrian poles of Gondwana indicating a clockwise rotation of about 38°. Since the age of rotation is not constrained, the authors assigned the rotation to the Andean deformation. However, they raised the possibility of interpreting the data as an Early Palaeozoic rotation affecting the whole Pampia block as part of final adjustments of Gondwana assembly (Fig. 8C). This possibility and its potential relations to the somewhat larger clockwise rotations found in the Famatina and the Eastern Puna awaits the acquisition of further palaeomagnetic data on Palaeozoic units in NW Argentina.

5 CONCLUSIONS

A palaeomagnetic study was carried out on Palaeozoic units exposed in central Famatina Range, NW Argentina. Samples of the Arenig-Llanvirn Molles Formation and the Cerro Morado Group were stepwise demagnetized by thermal and AF techniques. A pretectonic component of high temperature carried by magnetite was isolated and considered the primary magnetization in 14 sites. A palaeomagnetic pole was calculated from these results (MCM1) at 16.7°S 357.2°E ($A_{95} = 6.5°$, $N = 14$ sites), which is anomalous respect to the Early Ordovician reference pole for Gondwana, suggesting a clockwise rotation of ∼39°. However, this new pole is consistent with the previous palaeomagnetic results obtained on Early Ordovician rocks in the Famatina Ranges and the Eastern Puna Eruptive Belt of NW Argentina. Our results confirm the systematic large clockwise rotation that affected the Early Ordovician rocks of large areas of NW Argentina. Consistent Permian magnetizations found in Permian red beds and as remagnetization of Ordovician sediments exposed in the study area constrain the rotation as pre-Permian. Geological difficulties posed by the rotated terrane model suggest that alternative hypothesis as systematic crustal blocks rotations associated with subduction or collision processes along the SW margin of Gondwana or the rotation of a larger continental block ('Pampia') at the final stages of Gondwana assembly should be evaluated.

ACKNOWLEDGMENTS

This study was supported by CONICET (PIP 5783) and the University of Buenos Aires (UBACyT X262). Daniel Richarte fully collaborated during the fieldwork. F. Dávila kindly gave us access to his map of the study area. Mauro Spagnuolo is thanked for the help with the image treatment.

REFERENCES

Astini, R.A., 2003. The Ordovician Proto-Andean Basins, in *Ordovician Fossils of Argentina,* pp. 1–74, ed. Benedetto, J.L., Secretaría de Ciencia y Tecnología, Universidad Nacional de Córdoba, Argentina.

Astini, R.A. & Dávila, F.M., 2002. El Grupo Cerro Morado (Ordovícico Medio) en el Famatina (28°-29°S), Andes Centrales del oeste argentino, *Revista Geológica de Chile,* **29**(2), 151–165.

Astini, R.A. & Dávila, F.M., 2004. Ordovician back arc foreland and Ocloyic thrust belt development on the western Gondwana margin as a response to Precordillera terrane accretion, *Tectonics,* **23,** TC4008, doi:10.1029/2003TC001620.

Astini, R.A., Dávila, F., Collo, G., Ezpeleta, M., Dahlquist, J., Alasino, P. & Martina, F., in press. Geología de las Hojas Geológicas 2969–18 Famatina y 2969–24 Sañogasta, Provincia de la Rioja, *SEGEMAR,* Buenos Aires.

Benedetto, J.L., 1998. Early Palaeozoic brachiopods and associated shelly faunas from western Gondwana: their bearing on the geodynamic history of the pre-Andean margin, in *The Proto-Andean Margin of Gondwana,* pp. 57–83, eds Pankhurst, R.J. & Rapela, C.W., Geological Society of London Special Publication, 142.

Besse, J. & Courtillot, V., 2002. Apparent and true polar wander and the geometry of the geomagnetic field over the last 200 Myr, *J. geophys. Res.,* **107**(B11), doi:10.1029/2000JB000050.

Conti, C.M., Rapalini, A.E., Coira, B. & Koukharsky, M., 1996. Palaeomagnetic evidence of an early Palaeozoic rotated terrane in northwest Argentina: a clue for Gondwana-Laurentia interaction? *Geology,* **24,** 953–956.

Coughlin, T.J., O'Sullivan, P.B., Kohn, B.P. & Holcombe, R.J., 1998. Apatite fission-track thermochronology of the Sierras Pampeanas, central western Argentina: implications for the mechanism of plateau uplift in the Andes, *Geology,* **26**(11), 999–1002.

Dávila, F.M., 2003. Transecta estratigráfica-estructural a los 28°30′-28°45′ de latitud sur, Sierra de Famatina, Provincia de La Rioja, República Argentina, *PhD thesis,* University of Córdoba, Argentina, unpublished.

Dávila, F.M., Astini, R.A. & Schmidt, C.J., 2003. Unraveling 470 m.y. of shortening in the Central Andes and documentation of Type 0 superosed folding, *Geology,* **31**(3), 275–278.

Fisher, R.A., 1953. Dispersion of a sphere, *Proc. Roy. Soc. Lond., Ser. A*, **217**, 295–305.

Geeve, R., Schmidt, P., Roberts, J. & Lackie, M., 2001. Field tests and selective remagnetization in the New England Fold Belt, NSW, Australia, EGS XXVI General Assembly, Nice, France. SE14.02

Gradstein, F.M., Ogg, J.G. & Smith, A.G., 2004. *A Geologic Time Scale,* Cambridge University Press, Cambridge, UK, 589 pp.

Grunow, A.M., 1995. Implications for Gondwana of new Ordovician palaeomagnetic data from igneous rocks in southern Victoria Land, East Antarctica, *J. geophys. Res.,* **100**(B7), 12 589–12 603.

Harrington, H.J., 1957. Ordovician formations of Argentina, in *Ordovician Trilobites of Argentina,* pp.1–59, eds Harrington, H.J. & Leanza, A.F., University of Kansas Special Publication 1.

Hodych, J.P. & Buchan, K.L., 1994. Early Silurian palaeolatitude of the Springdale Group redbeds of central Newfoundland: a palaeomagnetic determination with a remanence anisotropy test for inclination error, *Geophys. J. Int.,* **117,** 640–652.

Kirschvink, J.L., 1980. The least-squares line and plane and the analysis of palaeomagnetic data, *Geophys. J. Roy. Astron. Soc.,* **67,** 699–718.

Llambías, E.J., Sato, A.M., Ortíz Suárez, A.E. & Prozzi, C., 1998. The granitoids of the Sierra de San Luis, in *The Proto-Andean Margin of Gondwana,* pp. 325–341, eds Pankhurst, R.J. & Rapela, C.W., Geological Society of London Special Publication, 142.

Lopez-Gamundi, O., Espejo, I., Conaghan, P. & Powell, C.Mc.A., 1994. Southern South America, in *Permian-Triassic Pangean Basins and Fold Belts along the Panthalassan margin of western Gondwanaland,* pp. 281–329, eds Veevers, J. & Powell, C.Mc.A., Geological Society of America, Memoir 184.

Lottes, A.L. & Rowley, D.B., 1990. Reconstruction of the Laurasian and Gondwanan segments of Permian Pangaea, *Palaeogeogr. Biogeogr. Geol. Soc. Mem.,* **12,** 383–395.

MacDonald, W.D., 1980. Net tectonic rotation, apparent tectonic rotation, and the structural tilt correction in palaeomagnetic studies, *J. geophys. Res.,* **85**(B7), 3659–3669.

Martínez, M.P. & Giménez, M.E., 2003. Fuerte anomalía gravimétrica residual positiva en el Sistema de Famatina y su relación con palaeosuturas. Explicaciones alternativas, *Revista de la Asociación Geológica Argentina,* **58**(2), 176–186.

McCabe, C., Jackson, M. & Saffer, B., 1989. Regional patterns of magnetite authigenesis in the Appalachian Basin: implications for the mechanism of Late Paleozoic remagnetization. *J. geophys. Res.* **94**(B8), 10 429–10 443.

McElhinny, M.W., 1964. Statistical significance of the fold test in palaeomagnetism, *Roy. Astr. Soc. Geophys. J.,* **8,** 338–340.

McFadden, P.L., 1990. A new fold test for palaeomagnetic studies, *Geophys. J. Int.,* **103,** 163–169.

McFadden, P.L. & Elhinny, M.W., 1990. Classification of the reversal test in palaeomagnetism, *Geophys. J. Int.,* **103,** 725–729.

Miller, H. & Sollner, F., 2005. The Famatina complex (NW Argentina): back-docking of an island arc or terrane accretion? Early Palaeozoic geodynamics at the Western Gondwana margin, in *Terrane Processes at the Margins of Gondwana,* pp. 241–256, eds. Vaughan, P.Leat & R.J. Pankhurst, Geological Society of London, Special Publication 246.

Pankhurst, R.J. & Rapela, C.W. (eds), 1998. *The Proto-Andean margin of Gondwana, Geol. Soc. London,* Special Publication, 142 pp.

Pankhurst, R.J., Rapela, C.W., Saavedra, J., Baldo, E., Dahlquist, J., Pascua, I. & Fanning, C.M., 1998. The Famatinian magmatic arc in the central Sierras Pampeanas: an Early to Mid-Ordovician continental arc on the Gondwana margin, in *The Proto-Andean Margin of Gondwana,* pp. 343–367, eds Pankhurst, R.J. & Rapela, C.W., Geological Society of London Special Publication, 142.

Pankhurst, R.J., Rapela, C.W. & Fanning, C.M., 2000. Age and origin of coeval TTG, I- and S-type granites in the Famatinian belt of NW Argentina, *Trans. Roy. Soc. Edinburgh, Earth Sci.,* **91**(1/2), 151–168.

Pankhurst, R.J., Rapela, C.W., Fanning, C.M. & Márquez, M., 2006. Gondwanide continental collision and the origin of Patagonia, *Earth Sci. Rev.,* **76,** 235–257.

Quenardelle, S.M. & Ramos, V.A., 1999. Ordovician Western Sierras Pampeanas magmatic belt: record of Precordillera acretion in Argentina, in *Laurentia-Gondwana connections before Pangea,* pp. 63–89, eds Ramos, V.A. & Keppie, J.D., Geological Society of America Special Paper 336.

Ramos, V.A., 1999. Rasgos estructurales del territorio argentino, in *Geología Argentina,* pp. 715–784, ed. Caminos, R., Anales No. 29 (24), Buenos Aires.

Ramos, V.A., 2004. Cuyania, an exotic block to Gondwana: review of a historical success and the present problems, *Gondwana Res.,* **7,** 1009–1026.

Ramos, V.A., Jordan, T.A., Allmendinger, R.W., Mpodozis, S., Kay, S.M., Cortés, J.M. & Palma, M.A., 1986. Palaeozoic terranes of the central Argentine–Chilean Andes, *Tectonics,* **5,** 855–880.

Rapalini, A.E., 2005. The accretionary history of Southern South America from the latest Proterozoic to the Late Palaeozoic: some Palaeomagnetic constraints, in *Terrane Processes at the Margins of Gondwana,* pp. 305–328, eds Vaughan, A., Leat, P. & Pankhurst, R.J., Geological Society of London, Special Publication 246.

Rapalini, A.E., 2006. New late Proterozoic palaeomagnetic pole for the Rio de la Plata craton: implications for Gondwana, *Precambrian Res.,* **147,** 223–233.

Rapalini, A.E. & Cingolani, C.A., 2004. First Late Ordovician paleomagnetic polo for the Cuyania (Precordillera) terrane of western Argentina: a microcontinent or a Laurentian Plateau? *Gondwana Res.,* **7**(4), 1089–1104.

Rapela, C.W., Pankhurst, R.J., Casquet, C., Baldo, E., Saavedra, J. & Galindo, C., 1998. Early evolution of the Proto-andean margin of South America, *Geology,* **26,** 707–710.

Rapalini, A.E., Fazzito, S.J. & Orué, D., 2006. A new Late Permian palaeomagnetic pole for stable South America: the Independencia Group, Eastern Paraguay, *Earth, Planet. Space,* **58**(10), 1247–1253.

Rapela, C.W., Pankhurst, R.J., Casquet, C., Fanning, C.M., Baldo, E.G., González-Casado, J.M., Galindo, C. & Dahlquist, J., 2007. The Rio de la Plata craton and the assembly of SW Gondwana. *Earth-Sci. Rev.,* **83**, 49–82.

Saavedra, J., Toselli, A., Rossi, J., Pellitero, E. & Durand, F., 1998. The Early Palaeozoic magmatic record of the Famatina System: a review, in *The Proto-Andean Margin of Gondwana,* pp. 283–295, eds Pankhurst, R.J. & Rapela, C.W., Geological Society of London Special Publication 142.

Spagnuolo, C.M., Rapalini, A.E. & Astini, R.A., 2007. Palaeogeographic and tectonic implications of the first palaeomagnetic results from the Middle-Late Cambrian Mesón Group: NW Argentina, *J. South Am. Earth Sci.*. doi:10.1016/j.jsames.2007.08.004

Tamrat, M. & Ernesto, M, 2006. Palaeomagnetic constraints on the age of the Botucatu Formation in Rio Grande do Sul, Southern Brazil, *Anais da Academia Brasileira de Ciências,* **78**(3), 591–605.

Thomas, W.A. & Astini, R.A., 2003. Ordovician accretion of the Argentine Precordillera terrane to Gondwana: a review, *J. South Am. Earth Sci.,* **16**(1), 67–79.

Turner, J.C.M., 1960. Estratigrafía del tramo medio de la Sierra del Famatina y adyacencias (La Rioja), *Boletín de la Academia Nacional de Ciencias,* **42,** 77–126.

Turner, J.C.M., 1964. Descripción geológica de la hoja 15c, Vinchina, *Boletín del Instituto Nacional de Geología y Minería,* 100pp, Argentina.

Turner, P., 1980. *Continental Red Beds*. Elsevier Scientific Pub. Co. Series Developments in sedimentology, 562 pp.

Valencio, D.A., Vilas, J.F. & Mendia, J.E., 1980. Palaeomagnetism and K-Ar ages of Lower Ordovician and Upper Silurian-Lower Devonian rocks from north-west Argentina, *Geophys. J. Roy. Astronom. Soc.,* **62,** 27–39.

Van Der Voo, R., 1990. Paleomagnetism of the Atlantic, Tethys and Iapetus Oceans, Cambridge University Press, Cambridge, UK, 411 pp.

Watson, G. and Enkin, R. 1993. The fold test in paleomagnetism as a parameter estimation problem. *Geophys. Res. Lett.,* **20**(19), 2135–2137.

Zambrano, O., Rapalini, A.E., Dávila, F.M., Astini, R.A. & Spagnuolo, C.M., in preparation. Magnetostratigraphy of the Neogene Del Crestón Formation, Famatina Ranges, NW Argentina: geochronologic and tectonic implication, *Tectonophysics*.